

DATA SCIENCE METHODOLOGY

Module 3 — Notes Summary

Compiled from my hand-written class notes

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1. What Is Methodology?

- **Definition:** a methodology is a system of methods used in a particular area of study.

REMEMBER: *Data Science is not just about models — success comes from following a structured process to solve real problems effectively.*

1.1 Why Methodology Matters

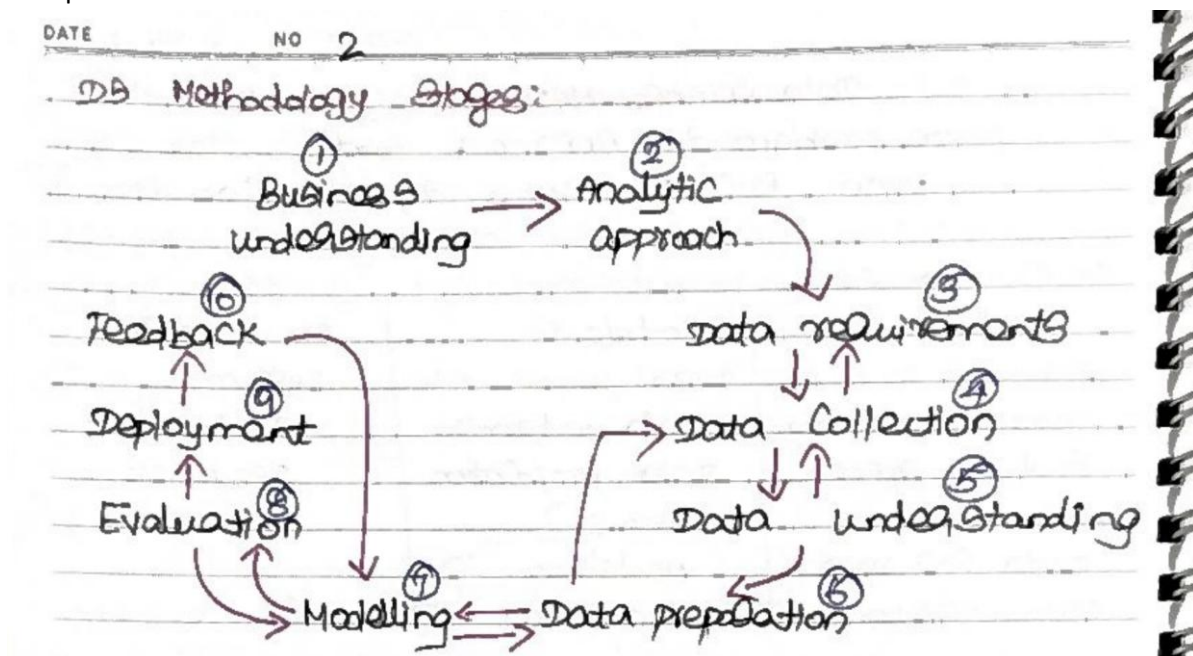
- People often jump straight into coding or modeling without clearly understanding the problem.
- A **methodology** acts like a **roadmap**. it helps you:
 - Ask the right questions
 - Use the right data
 - Apply the right technique
 - Avoid wasting time on the wrong solution

1.2 What “Data Science Methodology” Means

- A step-by-step framework that guides how you define a problem, work with data, build models, and turn results into real-world solutions.
- **Formalized by John Rollins**, who structured this approach into the 10-stage methodology covered below.

2. The 10-Stage Data Science Methodology

- The methodology is a cyclic, iterative process, not a one-way flow — later stages (e.g. Feedback) loop back into earlier ones.



From my hand-written notes — the 10 stages and how they cycle back into each other.

#	Stage	Key Question
1	Business Understanding	What is the problem you are trying to solve?
2	Analytic Approach	How can you use data to answer the business question?
3	Data Requirements	What data do you need to answer the question?
4	Data Collection	Where is the data sourced from, and how will you receive it?
5	Data Understanding	Does the data you collected represent the problem to be solved?
6	Data Preparation	What additional work is required to manipulate and work with the data?
7	Modeling	Which model is best suited for the problem?
8	Evaluation	Does the model answer the initial business question, or must the data be adjusted?
9	Deployment	Can you put the model into practice?
10	Feedback	Can you get constructive feedback from data-driven and the stakeholders to answer the business question?

3. Stage 1 — Business Understanding

- Don't start solving a problem until you fully understand what it really is.

KEY POINT: Most failures in Data Science don't happen because of bad models — they happen because people solve the wrong problem.

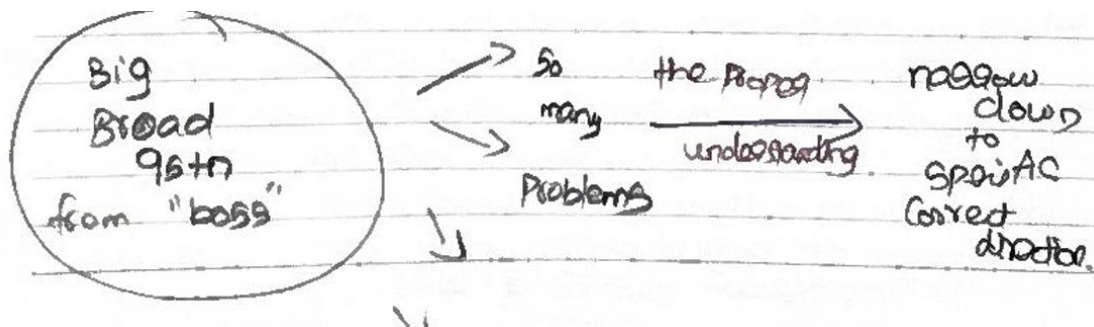
- You might build a technically perfect model, but it won't solve the actual business need.

3.1 Key Lessons

- Clarify the real goal
- Break it into clear objectives
- Align with stakeholders before touching any data

3.2 Example — Healthcare

- **Original question (broad, vague):** "How do we allocate the healthcare budget better?"
- **After business understanding (specific, actionable):** focus on patient re-admissions, especially for heart-failure patients.
- Takeaway: the exact insight (e.g. re-admission patterns) comes from data, but the decision to investigate re-admissions comes from business understanding.



4. Stage 2 — Analytic Approach

- Once you clearly understand the problem, you must choose the right type of analytical method — not just any model.

KEY IDEA: *Different questions require different types of analysis. The approach depends entirely on the nature of the question, not your favourite algorithm.*

4.1 Matching Outcome to Approach

Outcome You Need	Analytic Approach
Yes / No outcome	Classification
Probability of a future outcome	Prediction
Discover patterns or groups	Clustering
Understand relationships	Descriptive / Statistical analysis

4.2 Five Common Question Types

Question Type	Example Question	Approach	Techniques
Descriptive	What is the current status of our sales?	Descriptive analytics	Data aggregation, data mining, data visualization (dashboards, reports)
Diagnostic	Why did our sales decline last quarter?	Diagnostic analytics	Drill-down, data discovery, correlation analysis
Predictive	What is our sales forecast for next year ?	Predictive analytics	Regression analysis, time series forecasting, ML models
Prescriptive	What should we do to increase revenue?	Prescriptive analytics	Optimization, simulation (inventory, pricing, marketing)
Classification	Which category does this data point belong to?	Classification (supervised learning)	Decision trees, etc. (spam detection, image classification, disease diagnosis)

4.3 How the Case Study Applied It

- Business understanding defined the problem: predict patient readmission.
- Chosen analytic approach: classification problem (readmitted: Yes / No).
- Select classification algorithm - decision-tree classifier.

5. Stage 3 — Data Requirements

- The chosen analytic approach determines the data requirements — specifically, the data content and format needed.
- Before collecting any data, clearly define what data you actually need and in what form. Don't gather data randomly.

KEY POINT: *If you choose the wrong ingredients (data), even a perfect cooking process (modeling) will fail.*

5.1 What Data Requirements Includes

Data Content	Data Format	Data Source
What variables / features are needed? e.g. diagnosis, prescription, admissions	How should the data be structured? e.g. one row per patient, columns = features	Where will the data come from? e.g. hospital records, insurance database

5.2 What the Case Study Teaches

- Don't jump into data collection. First defined the target group: heart-failure patients with prior admission history, excluding confounding conditions.
- Thought about model requirements: since they chose a decision-tree, they needed structured, tabular data — one record per patient.
- Anticipated future steps: raw data = many records per patient, but the model needs one row per patient — so they planned an aggregation ("rolling up" the data) step for later, in Data Preparation.

SUMMARY: *Data Requirements = defining what data is needed, how it should look, and where it comes from — before collecting anything.*

6. Stage 4 — Data Collection

- Data Collection is not just "gather everything." It is an iterative process of collecting, checking, and adjusting data based on what's actually available and useful.

6.1 Reality

- Even after defining data requirements: some data is missing, some data is hard to access, some data is expensive or unavailable.
- So the data scientist must: collect what's possible → evaluate it → revise the plan if needed (collect more if needed).

KEY POINT: *Data Collection is flexible and iterative — not fixed. Not: define → collect → done. Instead: define → collect → assess → adjust → collect more (if needed). Not all data needs to be collected immediately.*

KEY TAKEAWAY: *Apply descriptive statistics and visualization techniques to assess content, quality and initial insights from collected data; identify gaps and determine whether more data is needed or existing data should be substituted.*

7. Stage 5 — Data Understanding

- About verifying whether the collected data truly reflects the real problem — and discovering issues before modeling.
- After collecting data, don't jump into modelling. Instead ask: "Does this data actually represent the problem I'm trying to solve?"
- This stage is mainly exploration and validation of data.

Technique	Purpose
Descriptive statistics (mean, median, min, max, std. dev.)	Helps understand the basic behaviour of variables
Relationships between variables — correlation analysis	Detects redundancy (highly correlated features)
Distribution analysis — histograms	Shows skewed data, outliers, need for transformation
Data quality checks — missing values, incorrect values (e.g. age = -99), inconsistent encoding	Very important — catches data issues early

DEEPER LESSON: *Your first version of data is almost always incomplete or misleading. Only by exploring it do you discover gaps, errors, and wrong assumptions.*

8. Stage 6 — Data Preparation

- Where raw data is cleaned, transformed and shaped into a form that models can actually use effectively.
- One of the most critical (and time-consuming) steps in the whole methodology.

Step	What It Does
Data Cleaning	Handle missing values, fix invalid entries, remove duplicates, improve data quality
Data Transformation	Normalize / scale values, convert formats, aggregate / restructure data, make data compatible with the model
Feature Engineering (heart of this stage)	Create new features from existing data using domain knowledge — e.g. combine multiple records into one meaningful feature, derive metrics like ratios or counts
Special Processing (e.g. text data)	Encode text into numeric form, group categories — no fixed time spent here as real-world data varies

KEY POINT: *Well-prepared data is the foundation of every successful data science model. Better data, better models.*

9. Stage 7 — Modeling

- Where you actually build models using the prepared data to answer the problem — training models to find patterns and make predictions.

9.1 Key Points

- **Two types of models:** descriptive models explain patterns; predictive models predict outcomes.
- **Use training data:** models are trained on historical data — since outcomes are known, you can check performance and tune (adjust) the model.
- **Experimentation is essential:** try different algorithms, adjust parameters, check which features matter.

- **Depends on earlier stages:** modeling success depends on correct problem understanding, the right analytic approach, and well-prepared data. If earlier stages are weak, the model will be too.
- **Continuous refinement:** models are not built once and done — you keep tweaking, re-evaluating and improving.

KEY POINT: *Modeling is not just applying algorithms. It is a process of experimentation and refinement to find the best solution.*

- Accuracy alone is misleading — you must also consider sensitivity (recall), specificity, and the cost of errors.

10. Stage 8 — Evaluation

- Checks whether the model actually solves the business problem, and helps choose the best version of the model.

10.1 Two Main Parts of Evaluation

- **Diagnostic Evaluation:** checks if the model is working properly. Uses metrics like sensitivity (true positive rate) and false positive rate. Helps identify where the model is weak and what needs tuning.
- **Statistical Evaluation:** ensures results are reliable and not random. Confirms the model is meaningful.

REMEMBER: *Evaluation is not just accuracy or AUC, etc. It is about: is this model solving the original business problem?*

- “Adjust data” can appear at this stage because: if the model fails, the issue may not be the model itself — it could be poor quality, missing important variables, or noisy / biased data.

SUMMARY: *Evaluation = selecting the best model based on performance, trade-offs and business needs.*

11. Stage 9 — Deployment

- About putting the model into real-world use so stakeholders can actually benefit from it.
- Building a model is not the end. A model is only valuable if: people can use it, it fits into real workflows, and it helps in decision-making.

11.1 What Happens in Deployment

- **Make the model usable:** convert the model into an application or tool; ensure it is easy to use (e.g. via browser, tablet).
- **Translate technical output:** the data science team builds the model; the business team explains the results — so non-technical users understand which is high-risk and what actions is to take.
- **Integrate into workflow:** real-time predictions, used during actual operations (e.g. patient discharge).

- **Involve stakeholders:** different people play different roles — business team, developers / IT team, end users (e.g. doctors). Collaboration is essential.

11.2 What Happened in the Case Study

- The model was turned into a risk-assessment tool.
- Doctors could see patient risk scores and take action before discharge.
- It was delivered via an easy-to-use application, and staff were trained to use it.

KEY POINT: *Deployment = turning a model into a usable system that supports real decisions.*

12. Stage 10 — Feedback

- About learning from real-world use of the model and continuously improving it.
- Even after deployment, the work is not finished.

12.1 What Happens in the Feedback Stage

- Monitor performance in real use
- Collect stakeholder feedback
- Refine the model
- Improve the overall system
- Repeat the cycle

12.2 What Happened in the Case Study

- Patients were tracked after the intervention.
- Readmission rates were compared before vs. after.
- Model impact was evaluated after one year.
- New improvements were identified: add drug data, refine the intervention process.
- Model was updated and redeployed.

KEY POINT: *Real-world performance is the ultimate test of a model, and continuous improvement is essential.*

- Feedback = monitoring, learning and improving the model after deployment.

13. CRISP-DM

- **CRISP-DM** = Cross-Industry Standard Process for Data Mining.
- It consolidates the steps outlined in the foundational 10-stage methodology into six stages:

• Business Understanding • Data Understanding	• Data Preparation • Modeling	• Evaluation • Deployment
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Like the 10-stage methodology, CRISP-DM is a cyclic, iterative process - it loops back rather than running once.